

# Faizan Khan

AI Engineer — Machine Learning Engineer — Generative AI Specialist

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## Professional Summary

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AI/ML Engineer and Computer Science student at UET Mardan with hands-on experience in **Machine Learning, Deep Learning, NLP, Generative AI, LLMs, and Agentic AI**. Proficient in **Python, PyTorch, Scikit-learn, LangChain, LangGraph, RAG, FAISS, and FastAPI**. Experienced in developing **AI agents, multi-agent systems, chatbots, RAG applications, and AI automation solutions**, with practical experience in model development, API integration, testing, and deployment.

## Core Skills

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### Programming & Frameworks

Python (Advanced), C++ (DSA & OOP), FastAPI, Scikit-learn, PyTorch, TensorFlow, LangChain, LangGraph, Hugging Face Transformers

### Artificial Intelligence & Machine Learning

Machine Learning, Deep Learning, NLP, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Multi-Agent Systems, Neural Networks, ANN Models, Model Optimization, Fine-Tuning, Prompt Engineering

### Data Structures & Algorithms

Graph Algorithms (A\*, Dijkstra), Greedy Algorithms, Trees & Linked Lists, Optimization Techniques, Human Coding

**Full Stack Development:** Developed AI-powered full-stack applications integrating machine learning (ML) models with React frontend and Python backend APIs

### Tools & Platforms

Git, GitHub, Jupyter Notebook, VS Code, OpenAI API, Groq API, Gemini API, FAISS Vector Databases, Data Visualization, Dashboard Development

### Soft Skills

Problem Solving, Critical Thinking, Research Mindset, Agile Development, Team Collaboration

## Work Experience

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### Generative AI Intern — Decode Labs

- Built LLM-powered chatbots, AI content, image generation & code review systems
- Worked with Python, GenAI, Prompt Engineering, Streamlit & REST APIs

### AI Agents & Automation Intern — AlgoHub

- Built AI agents and multi-agent automation workflows
- Recognized as **Top-Performing Intern**

## Machine Learning Fundamentals Intern — NeuroFive Solutions

- Applied Python, Machine Learning, data preprocessing & model evaluation

## Python & AI Developer

*Freelance / Academic Projects (3+ Years)*

- Developed end-to-end AI systems including data preprocessing, model training, optimization, and deployment
- Built NLP systems, chatbots, dashboards, and algorithm-driven intelligent applications
- Designed and implemented RAG-based AI systems using LangChain, LangGraph, and vector databases (FAISS)
- Applied Deep Learning and Machine Learning models to real-world predictive and classification problems
- Solved complex optimization problems using Data Structures and Algorithms (Graphs, Trees, Greedy methods)

## Projects

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### Generative AI & NLP Systems

- AI-Powered Chatbots using LLMs, LangChain & OpenAI
- AI Resume Screening & Ranking System
- OCR-Based Text Extraction System
- AI Study Assistant (PDFs, Notes, Voice & Quiz Generation)
- Mental Health Sentiment-Aware Chatbot
- UET Mardan University AI Chatbot
- AI Restaurant Chatbot (Ordering + Recommendation System)
- AI LinkedIn Reply Agent
- AI Twitter (X) Post Generator
- UPSC Essay Evaluation System (LLM-based)
- Multi-Agent Collaborative AI System

### RAG & LLM Systems

- PDF-Based Conversational RAG Chatbot
- High-Speed RAG Pipeline using Groq LLM
- Semantic Search Engine using Embeddings
- LangGraph Multi-Agent Conversational System
- Medical LLM Fine-Tuning using QLoRA (Unsloth)

- SupportAI — Customer Support Agent using RAG, LangGraph & FAISS
- Crop Knowledge Base & RAG Pipeline for Pakistan Agriculture
- Semantic Search & Document Retrieval using Embeddings

### **Machine Learning & Deep Learning**

- Car Price Prediction (Regression Models)
- House Price Prediction System
- Customer Churn Prediction using ANN
- Next-word prediction using Deep Learning
- Movie Recommendation System (Collaborative Filtering)
- Crop Recommendation System (Agricultural AI)
- Crop Disease Detection System using CNN

### **DSA & Algorithmic Systems**

- Route Optimization using Graph Algorithms (A\*, Dijkstra)
- Warehouse Robot Path Planning System
- Data Compression System (Huffman Coding)
- Task Scheduling System using Greedy Algorithms
- Student Management System using Trees & Linked Lists
- Smart Search Engine using Ranking Algorithms

### **Education**

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**University of Engineering and Technology Mardan**  
BS Computer Science (AI Specialization)

2022 – Present

**Board of Intermediate & Secondary Education**  
FSc Pre-Engineering

### **Achievements & Certifications**

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- Machine Learning Internship (3-Month Industry Training)
- Generative AI Certification – Arch Technology
- Machine Learning Certification – Developer Hub
- AI Agents: FastAPI Training
- web Certification - Funavry technology
- **NeuroFive Solutions** — Role: **Machine Learning Fundamentals**
- **AlgoHub** — Internship: **AI Agents & Automation**
- **Decode Labs** — Role: **Generative AI**